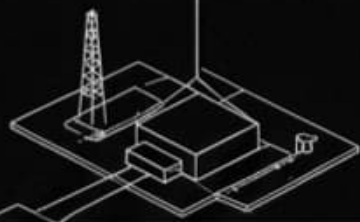




STUDENT-LED ATMOSPHERIC & NEAR-SPACE RESEARCH





A PROPOSED SOCIETY

From campus questions to atmospheric evidence.

IN-SPARX is a proposal for a student-led weather-balloon and atmospheric research society at MIT-WPU. It would give students a shared flight system for building payloads, measuring the atmosphere, recovering hardware, and turning observations into reproducible research.



PROPOSAL OWNER
Aryan Mishra

INSTITUTION
MIT-WPU

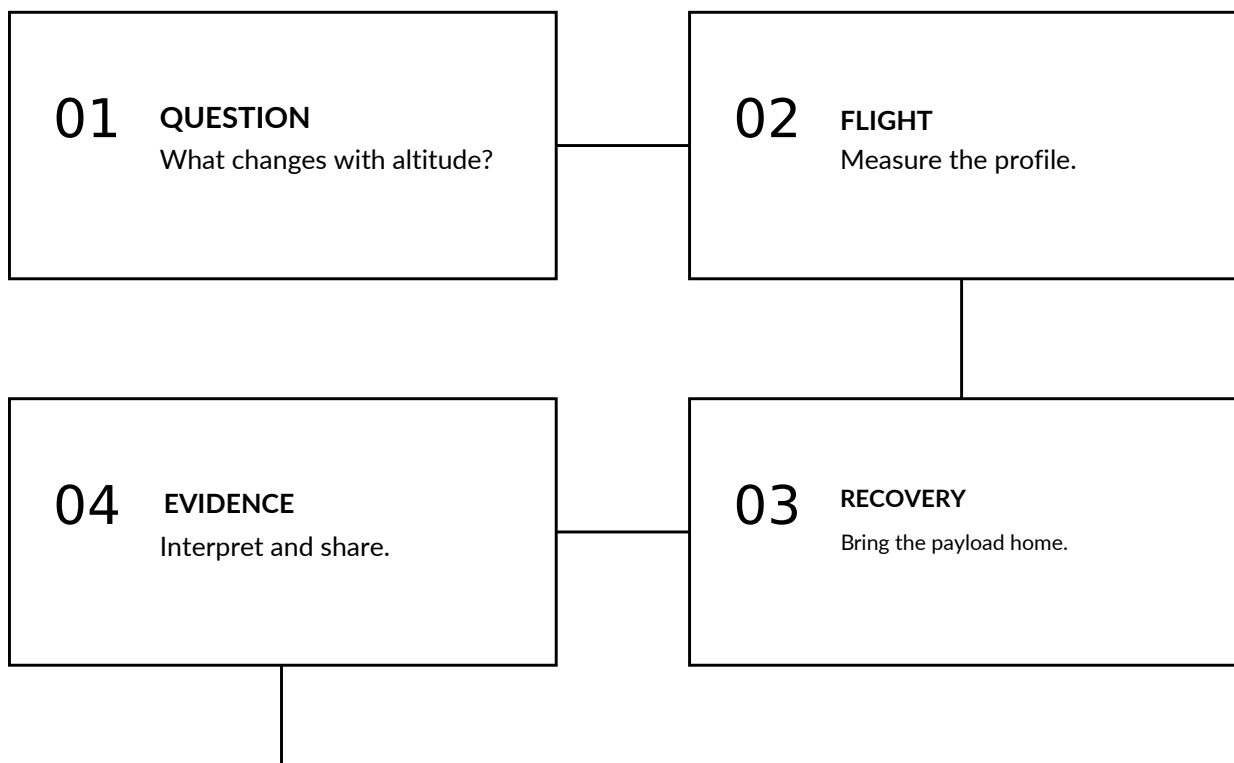
INSTRUMENTATION FOR NEAR-SPACE PHYSICS, AEROSPACE RESEARCH AND EXPLORATION

The atmosphere is the first laboratory above campus

Weather balloons lets students measure the atmosphere layer by layer instead of treating it as an invisible backdrop.

A radiosonde can measure pressure, temperature, relative humidity and GPS position as it rises. Tracking the changing position also reveals wind speed and direction. These are direct observations: a vertical record made by a student-built system rather than an abstract view from far above.

IN-SPARX would turn each observation into a repeatable research cycle: define the question, design the payload, test it, launch safely, recover it, analyze the data, and publish what was learned.



A flight is useful when the launch, measurements, recovery and analysis remain one connected system.

Why this idea should exist at MIT-WPU

IN-SPARX is not a one-off launch event. It is a student research society with faculty mentorship and institutional safety oversight.

01

ONE REAL SYSTEM

It connects aerospace, electronics, computer science, physics, environmental science and communication through one tangible flight system.

02

ONE VISIBLE PROJECT

It creates an evidence-driven platform that can attract students beyond a single department.

03

ONE COMPLETE CYCLE

Students practice requirements, design, integration, field operations, uncertainty, failure analysis and iteration.

04

ONE PUBLIC CONTRIBUTION

MIT-WPU gains a platform for student-led atmospheric observations, science communication and interdisciplinary collaboration.

INSTITUTIONAL POSTURE

Begin with a faculty-reviewed proposal, a defined safety process, and an interface to the institution's permissions and risk requirements. Operational details should be agreed before any flight activity.

Weather balloons answer questions satellites cannot answer alone

Satellites how broad-area patterns. Balloon soundings create a high-resolution vertical profile through the atmosphere.

TWO VIEWS. ONE ATMOSPHERE.

Pattern + profile + ground truth

SATELLITE

Wide-area view

Large spatial coverage and repeat observation.

BALLOON

Continuous vertical profile

Local, time-resolved measurements through the air column.

TOGETHER

Pattern + profile + ground truth

The balloon adds a measurement path a satellite cannot provide alone.

NOAA describes balloon flights as a three-dimensional cross-section used by forecasters, modelers and hurricane scientists. Balloon observations can improve forecast initialization and help ground-truth satellite data.

A typical radiosonde flight can last more than two hours and rise beyond 35 km, while experiencing temperatures near -90°C and pressure below 1% of sea-level pressure.

Why study the atmosphere?

Applied curiosity starts with a question that can be tested with data.

The atmosphere controls weather, air quality, aviation conditions, water cycles and the transport of heat and aerosols. Studying it above the surface shows how conditions change with altitude and how local observations connect to larger systems.



01

PROFILE THE AIR COLUMN

Measure vertical temperature, pressure and humidity profiles.

02

UNDERSTAND STABILITY

Study lapse rates, convection, inversions and boundary-layer behavior.

03

FOLLOW MOVEMENT

Study wind shear and how the atmosphere transports particles and moisture.

04

COMPARE MODELS

Set ground observations beside numerical weather models and satellite products.

05

BUILD LOCAL EVIDENCE

Use careful observations to improve environmental understanding around Pune.

Research themes IN-SPARX can grow into

A shared flight system can support different questions without becoming a collection of disconnected experiments.

01

THERMODYNAMICS

Lapse rates, stability, inversion layers and cloud-base conditions.

02

AIR QUALITY

Particles, gases and pollution gradients, with careful calibration.

03

URBAN BOUNDARY LAYERS

How Pune's built environment shapes near-surface conditions.

04

WIND AND DISPERSION

Wind profiles, turbulence proxies and aerosol movement.

05

RADIATION

UV, light and cosmic-ray proxy measurements with sensor shielding.

06

EARTH OBSERVATION

Horizon imaging, cloud structure and satellite comparison.

07

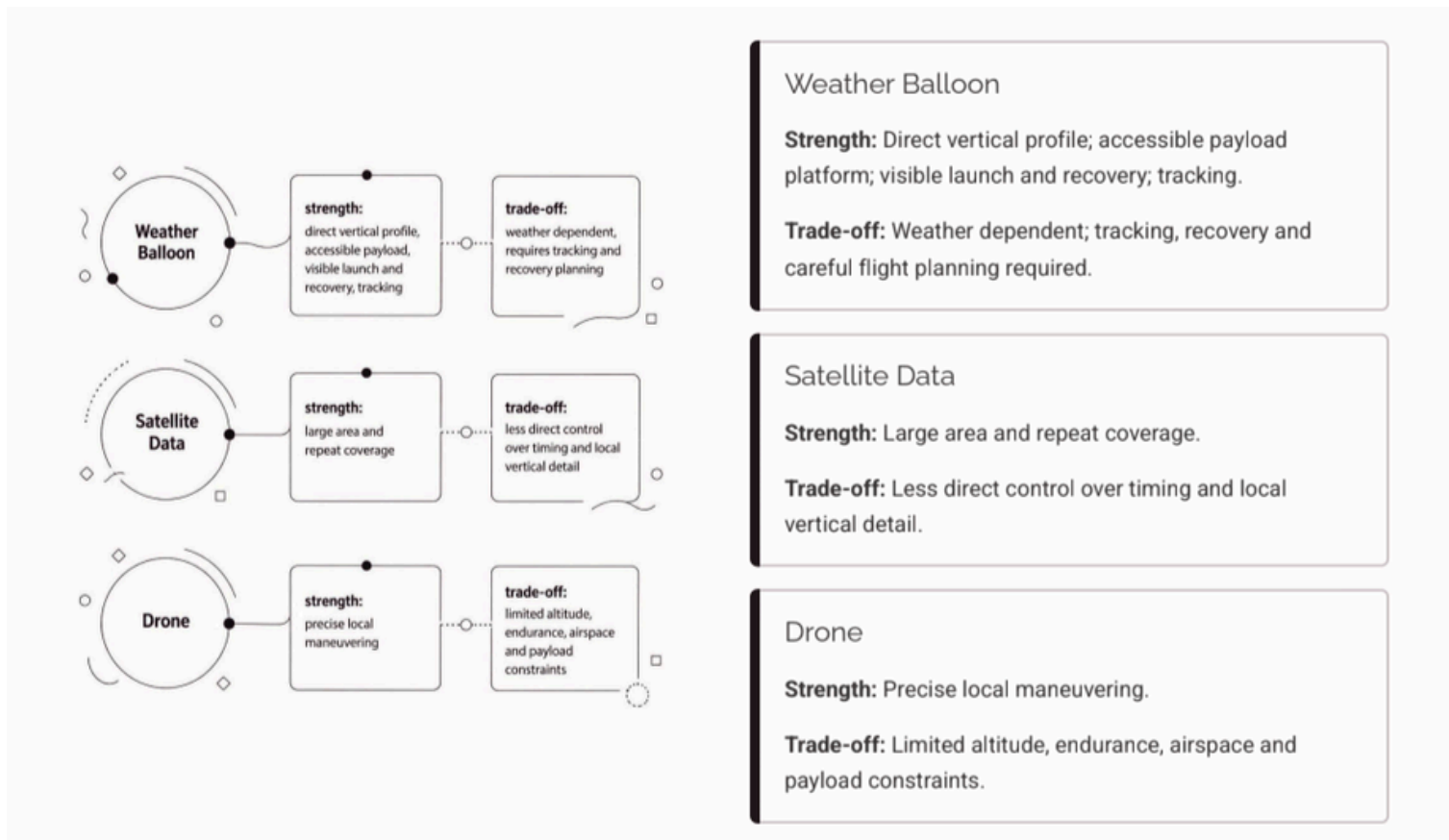
PAYLOAD ENGINEERING

Thermal design, power budgets, telemetry, recovery and reuse are research enablers in their own right.

Why choose weather balloons?

They combine reach, measurement quality and student ownership in one visible mission.

A balloon can carry a modular payload, deliver a continuous vertical profile, and make the invisible atmosphere tangible to the team and the public. The platform is accessible, but not casual: permissions, tracking, recovery and flight planning remain essential parts of the work.



DECISION PRINCIPLE

Choose the platform that preserves the question. For vertical atmospheric research, a balloon offers the strongest connection between the student payload and the measured air column.

Seven teams, one flight system

Each team owns a technical area, maintains documentation, and hands verified outputs to the next stage of the mission.



01	AVIONICS	Electrical architecture, flight computer, power and sensor interfaces.
02	TELEMETRY & RF	Communication link, ground receiver and tracking workflow.
03	INSTRUMENTATION	Sensor choice, calibration, shielding and uncertainty.
04	STRUCTURAL & THERMAL	Enclosure, mounts, insulation, parachute interface and protection.
05	SOFTWARE & DATA	Embedded code, dashboards, quality checks and analysis notebooks.
06	ATMOSPHERIC & SPACE SCIENCE	Questions, hypotheses, flight profiles and interpretation.
07	MISSION CONTROL	Integration, checklists, go/no-go decisions, permissions, launch operations, recovery and post-flight review.

The system works through interfaces: verified hardware, clean data, explicit decisions and documented handoffs.

What each team will do

The mission becomes executable whenever technical area has a clear owner, output and handoff.



01

BUILD THE ELECTRICAL NERVOUS SYSTEM

Avionics owns the flight computer, wiring, power budget, sensor buses, watchdogs and bench tests.

Deliverables: schematic, wiring harness, firmware interface and acceptance test.

02

MAKE THE PAYLOAD OBSERVABLE

Telemetry and RF owns the radiolink, antennas, groundreceiver, range tests and lost-link procedure.

03

PROTECT MEASUREMENT CREDIBILITY

Instrumentation owns requirements, calibrationrecords, shielding, sampling and uncertainty documentation.

04

KEEP HARDWARE WITHIN LIMITS

Structural and Thermal owns the enclosure, mounts, insulation, parachute interface, mass budget and mechanical tests.

05

TURN RAW SIGNALS INTO EVIDENCE

Software andData owns embedded logging, data schema, live display, quality checks and reproducible analysis.

06

DECIDE WHAT THE MISSION SHOULD LEARN

Atmospheric and Space Science owns hypotheses, variables, flight profile, comparisons and the final science report.

07

MAKE THE MISSION EXECUTABLE AND SAFE

Mission Control owns the integrated schedule, interface document, risk register, check lists, permissions tracker, launch-day roles, recovery plan and post-flight review.

Safety, permissions and recovery

Launch activity requires coordination with aviation authorities and nearby airports or air force stations, along with careful wind, cloud and recovery planning.

FLIGHT GATE

Permissions · hazard
register · tracking · recovery · go / no-go

Operational details should be agreed before any flight activity, with faculty mentorship and institutional safety oversight built into the process.



Launch safety controls

The flight system is designed around documented responsibility, permissions, tracking and responsible recovery.

- Appoint a faculty advisor and a student safety lead.
- Maintain a written hazard register and launch go/no-go checklist.
- Confirm required permissions and airspace coordination before every flight.
- Use redundant tracking and a recovery plan; the payload can land far from the release point.
- Prefer helium or approved lift systems, and document handling procedures.
- Design for safe descent, rounded edges, visibility and responsible recovery.



A phased path from society to flight program

01

Phase 1 • Foundation

recruit members, appoint mentors, define research questions, train teams and build a ground station.

02

Phase 2 • Ground validation

calibrate sensors, test telemetry, run environmental checks, simulate trajectories and rehearse recovery.

03

Phase 3 • First flight

fly a modest, well-instrumented payload with a narrow science question and clear success criteria.

04

Phase 4 • Analysis

publish a flight report, release cleaned data where appropriate, review failures and improve the design.

05

Phase 5 • Research cadence

run repeat flights that compare seasons, locations, payload configurations or hypotheses.

Visual: five-stage roadmap with a deliberate “learn → improve → repeat” loop.

What MIT-WPU gets in return

- A flagship interdisciplinary student initiative aligned with aerospace, science and engineering education.
- A visible platform for faculty-guided research and external collaboration.
- Public-facing demonstrations that make atmospheric science tangible to prospective students and visitors.
- Reusable infrastructure: a ground station, payload architecture, documentation, datasets and a trained student pipeline.
- A culture of responsible experimentation, where safety and evidence are treated as design requirements.

Approval should unlock mentorship, a small seed budget, workspace access, launch coordination support and a clear review pathway.

The decision requested

Approve IN-SPARX as a student-led MIT-WPU society exploring weather ballon

atmospheric instrumentation and near-space research under faculty and institutional oversight.

Start with a founding cohort, seven teams, a ground-testing semester and a permission-led first-flight proposal.

Let the first launch be the beginning of a repeatable research program, not the finish line.

OWNER Aryan Mishra



PART II

TIFR COLLABORATION

FROM STUDENT RESEARCH TO SCIENTIFIC BALLOONING



A proposed bridge between student-led research and India's scientific ballooning capability.

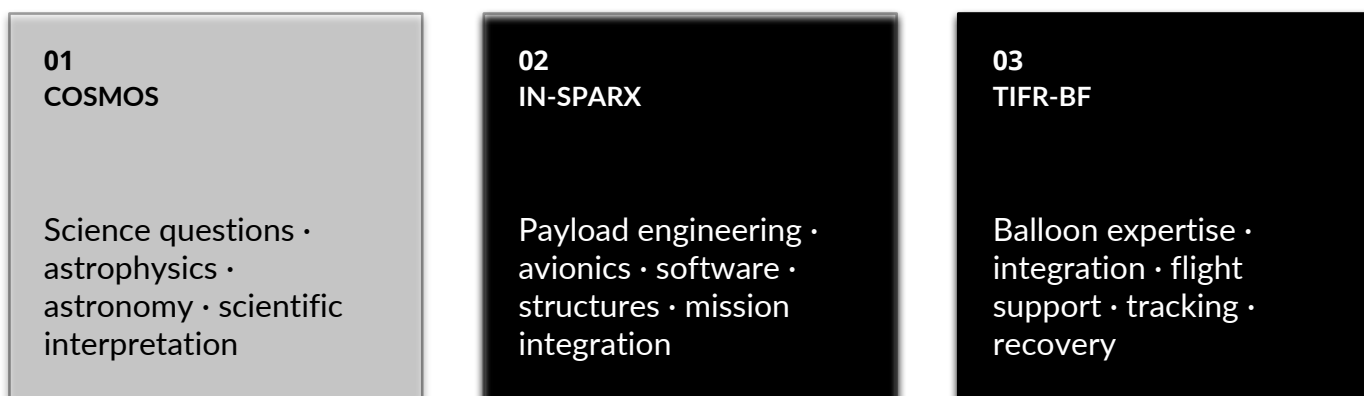
COSMOS + IN-SPARX · MIT-WPU · TIFR BALLOON FACILITY, HYDERABAD

01 / COLLABORATION

ONE FLIGHT SYSTEM, LED BY A BRILLIANT COLLABORATION

A proposed bridge between student-led research and India's scientific ballooning capability.

IN-SPARX is designed as a student-led weather-balloon and atmospheric research society at MIT- WPU. COSMOS brings a complementary science and astronomy community around the project. TIFR Balloon Facility can provide the expert technical and operational interface that turns a student-built concept into a rigorously reviewed balloon experiment.



THE INTENDED INTERFACE

TIFR expertise would sit alongside the student teams—not replace them. Students own the questions, engineering and analysis while experienced balloon specialists review, guide and enable the parts that require institutional capability.

CONCEPTUAL COLLABORATION MODEL · TIFR CAPABILITY REFERENCES: ANNUAL REPORTS / SELF STUDY REPORT

02 / WHY TIFR-BF

WHY TIFR-BF IS THE RIGHT INTERFACE

The facility's documented capability spans the balloon mission chain, not just the balloon itself.

01 BALLOON DESIGN & FABRICATION

Documented in-house design and fabrication of scientific and tethered balloons for different payload classes and mission needs.

02 PAYLOAD INTEGRATION

Experience integrating payload packages with telemetry, telecommand and other flight instrumentation.

03 FLIGHT OPERATIONS

Launch support, tracking, data collection, flight control and payload recovery.

04 INSTRUMENTATION

Telemetry, GPS, transponder and antenna-related support systems for balloon missions.

05 RESEARCH PLATFORM

Scientific ballooning as a low-cost, frequent-access platform for atmospheric, space and astrophysical research.

06 USER-FOCUSED MODEL

Support for user scientists from national and international institutions through balloon R&D and operations.

SOURCE FRAME

TIFR Research Facilities / Annual Reports · NAAC Self Study Report · Scientific Talks (Balloon Facility)

03 / VALUE TO TIFR

POTENTIAL VALUE TO TIFR

A student-facing pipeline can complement the facility's research, engineering and outreach ecosystem.

01**REPEATABLE STUDENT PAYLOAD PIPELINE**

Small, modular experiments documented from question → design → bench validation → flight proposal → analysis.

02**NEW INSTRUMENTATION CONCEPTS**

A structured environment for low-cost sensors, data systems, payload electronics and measurement concepts.

03**INTERDISCIPLINARY RESEARCH COMMUNITY**

Physics, aerospace, electronics, software and atmospheric science students become a trained pool.

04**OUTREACH & SCIENCE COMMUNICATION**

Joint demonstrations, student talks and carefully reviewed public-facing results can broaden ballooning visibility.

THE PROPOSITION

The goal is not to ask TIFR to run a student club. It is to create a defined interface where TIFR expertise can help a university team develop experiments to a standard suitable for serious scientific ballooning.

Potential benefits are a proposal, not commitments by TIFR.

04 / MIT-WPU GAINS

WHAT MIT-WPU GAINS

The collaboration turns a student society into a structured research-and-engineering pathway.

01
TECHNICAL MENTORSHIP

Students learn how balloon experiments are specified, reviewed, integrated and tested.

02
REAL MISSION CONSTRAINTS

Teams work with mass, power, telemetry, measurement quality, tracking and recovery.

03
RESEARCH DATA

Where appropriate and approved, flight data can become material for student analysis, reports and future experiments.

04
FACULTY + STUDENT PIPELINE

A recurring project can connect faculty guidance with students from physics, aerospace, electronics and computing.

05
RESEARCH CULTURE

Documentation, uncertainty, safety and reproducibility become part of the science.

06
LONG-TERM CAPABILITY

Each mission leaves trained people, documentation, payload architecture and lessons for the next cohort.

OUTCOME A successful partnership would be measured by student capability and research quality —not simply by whether a balloon is launched.

05 / TIFR ROLE

HOW TIFR CAN HELP IN-SPARX

A possible role at every major transition from student concept to flight-ready experiment.

01	TECHNICAL REVIEW	Review the science question, payload concept, interfaces and constraints.
02	DESIGN GUIDANCE	Advise on payload architecture, mass/power budgets, mechanical interfaces and measurement requirements.
03	INTEGRATION & TEST	Guide appropriate bench, environmental and communication tests before any flight proposal.
04	FLIGHT SYSTEM INTERFACE	Advise on telemetry, tracking, flight-control interfaces and operational documentation.
05	MISSION SUPPORT	Where approved and feasible, provide an institutional pathway for launch, tracking, data collection and recovery.
06	POST-FLIGHT SCIENCE	Help students understand data quality, calibration, uncertainty, analysis and technical reporting.

Specific support would be defined jointly and remain subject to TIFR policies, capacity, safety review and institutional approvals.

06 / REQUEST

WHAT WE ARE REQUESTING FROM TIFR

A focused list of support that would help IN-SPARX become technically credible and institution-ready.

01
ATECHNICAL POINT OF CONTACT

A scientist/engineer or designated interface for periodic technical review.

02
REQUIREMENTS REVIEW

Feedback on payload requirements, interfaces, documentation and test philosophy.

03
PAYLOAD ENGINEERING GUIDANCE

Advice on mechanical, electrical, thermal and instrumentation integration.

04
TELEMETRY & TRACKING GUIDANCE

Guidance on communications, tracking and data architecture appropriate to balloon missions.

05
FACILITY / EXPERTISE ACCESS

Access to relevant expertise, demonstrations or facilities where TIFR policies permit.

06
FLIGHT PATHWAY

Advice on whether and how a suitable IN-SPARX payload could enter a future balloon experiment pathway.

07
STUDENT TRAINING

Technical talks, facility exposure or supervised learning opportunities for selected student teams.

08
RESEARCH COLLABORATION

Guidance on data quality, scientific analysis, reporting and—where appropriate—future joint research.

Requested support is subject to TIFR scientific priorities, operational capacity, safety requirements and institutional permissions.

07 / PARTNERSHIP

A STAGED PARTNERSHIP

Start with technical exchange. Earn the right to propose a flight through evidence and preparation.

01**INTRODUCE**

TIFR reviews IN-SPARX, its student teams and initial science questions.

02**DEFINE**

Jointly identify a small, technically realistic first payload concept.

03**BUILD**

Students develop the payload under faculty oversight, with TIFR technical feedback.

04**VALIDATE**

Complete documented ground testing and close identified technical gaps.

05**PROPOSE**

Submit a formal experiment / flight proposal through the appropriate TIFR process.

06**LEARN**

If approved and flown, analyze the data, document lessons and define the next experiment.

THE PRINCIPLE

IN-SPARX should arrive at TIFR not as an unfinished idea asking for a launch, but as a documented student research program asking for expert review and a pathway to scientific ballooning.

A collaboration can begin well before the first flight.